Cache in Storage Systems (Read & Write Cache)

1. 10-minute overview for Block, File, and Object storage environments.

What is Cache?

1. High-speed temporary storage layer (RAM/SSD)
2. Stores frequently accessed data
3. Reduces latency and improves performance
4. Acts as a bridge between application and storage

Why Cache Matters

1. Disks and network storage are slower than memory
2. Cache reduces read/write latency
3. Improves throughput and responsiveness
4. Critical for scaling storage workloads

Read Cache

1. Stores frequently accessed read data
2. Serves repeated requests quickly
3. Works using cache hit and cache miss

Write Cache

1. Improves write performance
2. Data written to cache first
3. Two strategies: Write-through and Write-back

Write-Through

1. Data written to cache and storage together
2. Safer and consistent
3. Slightly slower writes

Write-Back

1. Data written to cache first
2. Flushed to storage later
3. Faster but risk if cache fails

Dirty Data

1. Data present in cache but not yet written to storage
2. Eventually flushed to disk

Latency Comparison

1. RAM: ~50–100 nanoseconds
2. SSD: ~100 microseconds
3. HDD: ~5–10 milliseconds
4. RAM is thousands of times faster than disk

Architecture Flow (Conceptual)

1. Application → Cache → Storage
2. Reads: Check cache first, then storage on miss
3. Writes: Go to cache first, then flushed to storage